Chapter ~ 7

Alcohols, Phenols & Ethers.

⟹ Alcohols — Classification and Nomenclature.

$$R-H \xrightarrow{+OH \quad -H} R-OH$$

Classification by degree of carbon bearing.

Type	Example	Structure
1° (Primary)	Ethanol	CH_3-CH_2-OH
2° (Secondary)	Isopropanol (propan-2-ol)	$CH_3-CH(OH)-CH_3$.
3° (tertiary)	tert - Butanol	$CH_3-\underset{CH_3}{\overset{OH}{C}}-CH_3$.

General Formula : Saturated acyclic monohydric alcohols

$C_nH_{2n+1}OH$ or $(C_nH_{2n+2}O)$

⟹ Difference between 1°, 2° and 3° Alcohols.

	Property	1° Alcohol	2° Alcohol	3° Alcohol.
i]	Structure	-OH attatched to only one other C	-OH carbon attatched to two other C	-OH carbon attatched to three other C.
ii]	Boiling point	Highest	Intermediate	Lowest.
iii]	Oxidation	→ Carboxylic acid	→ Ketone	→ No reaction.
iv]	Dehydration ease	Slowest	faster	fastest.
v]	Lucas reagent	No turbity	Turbid in mins	Turbid Instantly
vi]	SN_2 substitution	Fastest	Slower	Slowest

__/2/105

⟹ Classification by number of $-OH$ groups.

i) Monohydric Alcohol — one $-OH$ groups
(eg Ethanol)

ii) Dihydric Alcohol — two $-OH$ gps : $OH-CH_2-CH_2-OH$
(ethylene glycol)

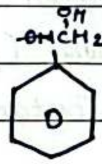
iii) Polyhydric Alcohol — three $-OH$ groups :

(Glycerol) $OH-CH_2-\overset{OH}{\overset{|}{CH}}-CH_2OH$

⟹ Special Structure Classes.

- Allylic Alcohol : $-OH$ on carbon adjacent to a $C=C$ double bond.

$CH_2=CH-CH_2-OH$. (prop-2-en-1-ol).

- Benzylic Alcohol :  (Benzyl Alcohol)

⟹ Methods of Preparation of Alcohols.

(a) From Alkyl Halides. — Hydrolysis

$$R-X + \begin{matrix} aq.\ KOH \\ aq.\ NaOH \\ moist\ AgOH \end{matrix} \longrightarrow R-OH + \begin{matrix} K-X \\ Na-X \\ Ag-X \end{matrix}$$

Example : ①

$$C_2H_5Br \xrightarrow{aq.\ KOH} C_2H_5OH + KBr.$$

Example : ②

$$C_2H_5Br \xrightarrow{alc\ KOH} CH_2=CH_2 + KBr + H_2O$$

/3/106

Relative Tendency

- SN2 Substitution : 1° Alc > 2° Alc > 3° Alc
- Elimination (Alkene) : 3° Alc > 2° Alc > 1° Alc

$$CH_3-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-Br \xrightarrow{NaOH} CH_2=\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}} + NaBr + H_2O.$$

(b) From Alkenes. — (Hydration).

i) Acid catalysed direct hydration

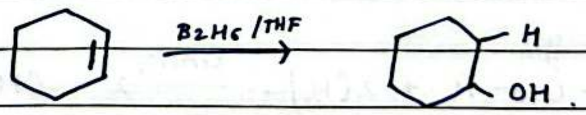
Addition of H-OH needs an acidic catalyst.

- $CH_2=CH_2 + H_2O \longrightarrow$ No reaction (uncatalysed)
- $CH_3-CH=CH_2 \xrightarrow[H_2O]{dil\ H_2SO_4} CH_3-\underset{OH}{\underset{|}{CH}}-CH_3$ Markownikoff's Rule.

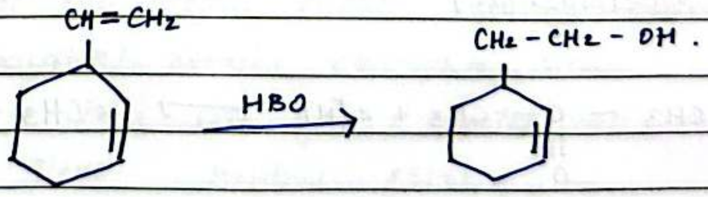
ii) Hydroboration — oxidation (anti-markownikoff, indirect hydration).

$$H_3C-CH=CH_2 \xrightarrow[2)\ H_2O_2\,/\,OHO]{1)\ B_2H_6\,/\,THF} CH_3-\underset{H}{\underset{|}{CH}}-\underset{OH}{\underset{|}{CH_2}}.$$

Example : 1



Example : 2



- Anti-markownikoff Rule
- Rate $\propto$ 1 / Steric hindrance
- BH_3 is unstable and becomes B_2H_6

iii) Oxymercuration - Demercuration (Markovnikov, addition of H_2O)

$$CH_3-CH=CH_2 \xrightarrow[\text{② } H_2O \text{ ③ } NaBH_4]{\text{① } (CH_3COO)_2Hg} \underset{\substack{| \\ OH}}{CH_3-CH}-\underset{\substack{| \\ H}}{CH_2}.$$

(c) From Grignards Reagent.

$$R-MgX + \begin{matrix} \text{① } R_2CHO \\ \text{② } R_2R_3C=O \end{matrix} \longrightarrow \text{addition pdt} \longrightarrow \text{alcohol}$$

Example : ① $HCHO + R-MgX \longrightarrow 1^\circ$ Alcohol.
(formaldehyde)

② $R_2CHO + R-MgX \longrightarrow 2^\circ$ Alcohol
(other aldehydes)

③ $R_2R_3C=O + R-MgX \longrightarrow 3^\circ$ Alcohol

$$CH_3-CH_2-O-H+CH_3MgBr \longrightarrow CH_4 + Mg(OCH_2CH_3)Br.$$

(d) Reduction of Carbonyl compounds.

Reagents : H_2/Ni, H_2/Pt, H_2/Pd or $LiAlH_4$ / $NaBH_4$

- $H-\overset{\overset{O}{||}}{C}-H + 2[H] \xrightarrow{H_2/Ni \text{ or } H_2/Pt} CH_3-OH$
  (formaldehyde) (methanol) 1°

- $CH_3-\overset{\overset{O}{||}}{C}-H + 2[H] \xrightarrow{LiAlH_4} CH_3-CH_2-OH$
  (acetaldehyde) (ethanol) 1°

- $CH_3-\underset{\underset{O}{||}}{C}-CH_3 + 2[H] \longrightarrow CH_3-\underset{\substack{| \\ OH}}{CH}-CH_3$
  (propanol) 2°

Note: 3° Alcohols can not be prepared by this

(e) From Primary Amines.

$$R-NH_2 \xrightarrow{NaNO_2+HCl} R-N_2^+Cl^- \xrightarrow[warm]{H_2O} R-OH + N_2 + HCl.$$

(f) Manufacture of Methanol and Ethanol.

i) Methanol – Bosch Process

$$CO + 2H_2 \xrightarrow[300\ atm,\ 673\ K]{ZnO/Cr_2O_3} CH_3OH.$$

ii) Ethanol – Fermentation of Carbohydrates.

$$(C_6H_{10}O_5)_n \xrightarrow[diastase]{+nH_2O} n\,C_{12}H_{22}O_{11} \quad (Starch \rightarrow maltose)$$

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{maltase} 2C_6H_{12}O_6 \quad (Maltose \rightarrow Glucose).$$

$$C_6H_{12}O_6 \xrightarrow{zymase} 2C_2H_5OH + 2CO_2$$

⟹ Chemical Properties of Alcohols.

(a) Esterification

(ethylene).

$$OHCH_2-CH_2OH + 2CH_3COOH \xrightarrow[H_2SO_4]{conc} CH_3COO-CH_2-CH_2-OOCCH_3 + 2H_2O$$

Mechanism : (Fischer esterification)

1. Protonation of carbonyl oxygen makes carbonyl carbon electrophile
2. the alcohol (nucleophile) attacks electrophile carbon
3. Proton transfer, then loss of a water molecule → carbonyl.
4. Deprotonation gives neutral ester.

$$R-COOH + R_2-OH \rightleftharpoons R-COOR_2 + H_2O$$

(b) Dehydration of Alcohols

$$R-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3 \xrightarrow[H_3PO_4 \text{ or } KHSO_4]{conc.\ H_2SO_4} R-CH=CH_2 + H_2O$$

Mechanism

1. (Protonation) $R-\overset{OH}{CH}-CH_3 \longrightarrow R-\overset{OH_2}{CH}-CH_3$
2. (loss of water) $R-\overset{+OH_2}{CH}-CH_3 \xrightarrow{-H_2O} R-CH-CH_3$
3. (loss of B-H alkene) $R-CH-CH_3 \longrightarrow R-CH=CH_2$

⟹ Saytzeff's Rule

Example : (1)

$$CH_3-\overset{OH}{CH}-CH_3 \xrightarrow{conc.\ H_2SO_4} CH_2=CH-CH_3 \text{ (propene)}$$

(2)

$$CH_3-\overset{OH}{CH}-C_2H_5 \longrightarrow CH_3-CH=CH-CH_3 \text{ (major)}$$
$$CH_2=CH-C_2H_5 \text{ (minor)}$$

⟹ Carbocation rearrangement during dehydration.
When It is not stable It shifts to 1,2 hydride shift or 1,2 methyl shift

Example : (1) 1,2 hydride shift ($1^\circ - 2^\circ$)

$$CH_3-CH_2-CH_2-CH_2 \xrightarrow{1,2\ hydride} CH_3-CH-CH_2-CH_3$$

Example : (2) 1,2 methyl shift

$$CH_3-\underset{CH_3}{\overset{CH_3}{C}}-CH_2-OH \xrightarrow{conc.\ H_2SO_4} CH_3-\underset{CH_3}{\overset{CH_3}{C}}-CH_2 \quad (1^\circ)$$

$$CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2 \xrightarrow{\text{1,2 methyl shift}} CH_3-\overset{\overset{CH_3}{|}}{C}-CH_2-CH_3 \quad (3^\circ)$$

Important Exception — where no carbocation occurs

" If the dehydrating agent is Al_2O_3, P_2O_5 or $POCl_3$ instead of conc. H_2SO_4 no carbocation or rearrangement occurs, only saytzeff rule

Example : (1)

$$CH_3-CH_2-CH_2-CH_2-OH \xrightarrow{Al_2O_3} CH_3-CH_2-CH=CH_2$$

(C) Oxidation of Alcohols

$$\text{Alcohols} \xrightarrow{[O]} \text{Aldehyde / Ketone} \xrightarrow{[O]} \text{Carboxylic Acid}$$

Product depends on

i) Type of Alcohol ii) type of reagent.

Example : (1) with Acidic $K_2Cr_2O_7$ or $KMnO_4$ and Red hot tube (Cu)

Alcohol type	$KMnO_4$ Acidic $K_2Cr_2O_7$	Red hot tube
1° ($R-CH_2-OH$)	carboxylic Acid $R-COOH$	Aldehyde $R-CHO$
2° ($R-\underset{OH}{CH}-R$)	Ketone $R-CO-R$	Ketone $R-CO-R$
3° Alcohol	No reaction	Alkene (loss of H_2O).

/8/111

Imp Example : (2) Pyridium Chloro Chromate
1° Alcohol to aldehydes only

Example : (3) Jones Reagent : H_2CrO_4

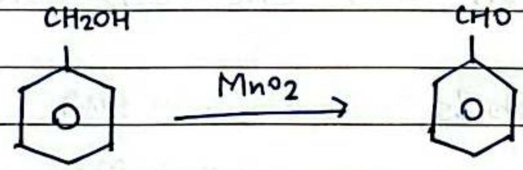
Example : (4) Collins Reagent : CrO_3 : pyridine : CH_2Cl_2
1 : 2 : 1
1° Alcohol to aldehydes

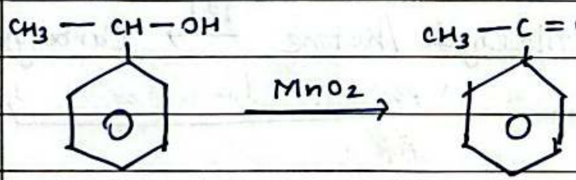
Example : (5) Sarrets Reagent :
1° Alcohol to aldehydes

Example : (6) NBS : allylic / Benzylic bromination.

Example : (7) Haloform

Example : (8) MnO_2 : Only oxidises Benzylic / Allylic Alcohols.


(Benzaldehyde)


(Acetophenone)

Example : (9) OPR Oxidation (Oppenauer's oxidation)
Only works on 2° Alcohols

$$CH_3 - \underset{\displaystyle \overset{|}{OH}}{CH} - CH_3 \xrightarrow{Al(OCMe_3)_3} CH_3 - \overset{\displaystyle O \atop \displaystyle \|}{C} - CH_3$$

(Ketone)

/9/112

(d) Reduction of Carboxylic Acids

RCOOH (acid) / RCOX (acid halide) / $RCOOR_2$ (ester)

1) Carboxylic Acid :

$$R-COOH \xrightarrow{LiAlH_4} R-CH_2OH$$

$$R-COOH \xrightarrow{NaBH_4} \text{No Reaction.}$$

2) Ester :

$$R-COOR_2 \xrightarrow{LiAlH_4} R-CH_2OH + R_2OH$$

$$R-COOR_2 \xrightarrow{NaBH_4} \text{No Reaction}$$

Example: ① $CH_3-COOCH_3 \xrightarrow{LiAlH_4} CH_3-CH_2-OH + CH_3OH.$

3) Acid Chloride :

$$R-COCl \xrightarrow{LiAlH_4} R-CH_2OH + HCl$$

$$R-COCl \xrightarrow{NaBH_4} \text{No Reaction.}$$

$LiAlH_4$ reduces All three.

$NaBH_4$ reduces none $\longrightarrow$ reserved for aldehyde/ketone.

(e) Acidic Nature of Alcohols.

$$R-OH \rightleftharpoons R-O^- + H^+ \quad \text{(alkoxide ion)}$$

Alcohols are less acidic than water.
and much less acidic than Phenols.

Acidity order : $1^\circ_{alc} > 2^\circ_{alc} > 3^\circ alc.$

(f) Reaction with Sodium.

$$2R-OH + 2Na \longrightarrow 2R-ONa + H_2$$

(sodium alkoxide)

Brisk effervescence.

(g) Reaction with hydrogen halides (HX)

$$R-OH + HX \longrightarrow R-X + H_2O.$$

Reactivity of : HX : HI > HBr > HCl.
Reactivity of OH : $3^\circ > 2^\circ > 1^\circ$.

(h) Reaction with PCl_3, PCl_5 and $SOCl_2$.

i) PCl_3 : $3R-OH + PCl_3 \longrightarrow 3R-Cl + H_3PO_3$

ii) PCl_5 : $R-OH + PCl_5 \longrightarrow R-Cl + POCl_3 + HCl$

iii) $SOCl_2$: $R-OH + SOCl_2 \longrightarrow R-Cl + SO_2 + HCl$
(Darzens process).

(i) Reaction with Acid Chlorides and Acid Anhydrides

i) $R-OH + R_2COCl \longrightarrow R_2COOR + HCl$
(ester).

ii) $R-OH + (R_2CO)_2O \longrightarrow R_2COOR + R_2COOH$
(ester)

⟹ Uses of Alcohols.

- Methanol : solvent, denaturant for ethanol possible alternative fuel, additive.
- Ethanol : Solvent, in beverages, as an antiseptic, as a fuel
- Generally, preparation of dyes, drugs, Perfumes, preservative.

⟹ Conversion of one alcohol into another.

$C_2H_5OH \xrightarrow{PCl_5} C_2H_5Cl \xrightarrow{Mg\ dry\ ether} C_2H_5MgCl \xrightarrow{HCHO}$ 1° Alcohol $\xrightarrow{aldehyde/ketone}$ 2° or 3° Alcohol.

⟹ Distinction between 1° and 2° and 3° Alcohols

Lucas's Test.

Lucas's reagent : anhydrous $ZnCl_2$ + conc. HCl.

Principle : Alcohol ⟶ Alkyl Chloride

$R-OH + HCl \xrightarrow{Lucas's} R-Cl + H_2O$

(turbidity)

Time of turbidity : 1° > 2° > 3°

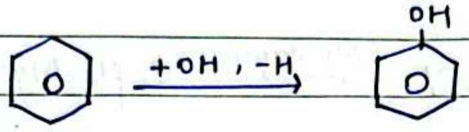
no turbidity

fastest.

⟹

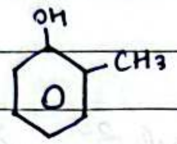
/12/115

⟹ Phenols (Classification, Nomenclature).

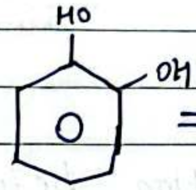


(Phenol)

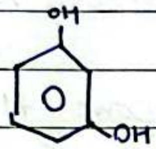
- Common Polyhydric Phenols



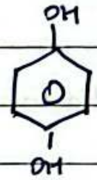
⟹ o- Cresol



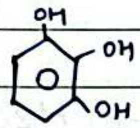
⟹ Catechol



⟹ Resorcinol



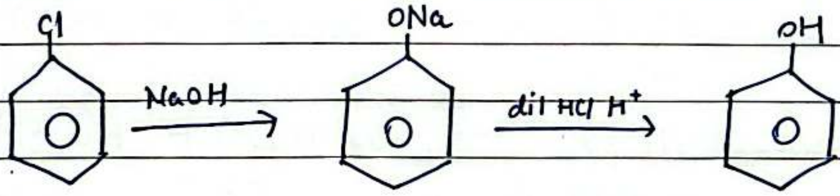
⟹ Quinol



⟹ Pyrogallol.

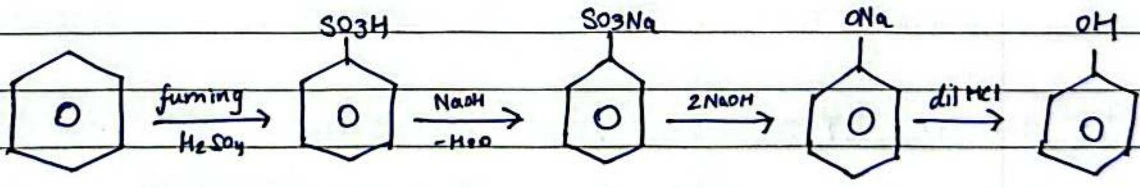
⟹ Preparation of Phenols.

(a) From Haloarenes — Dow's Process



Chlorobenzene   Sodium Phenoxide   Phenol.

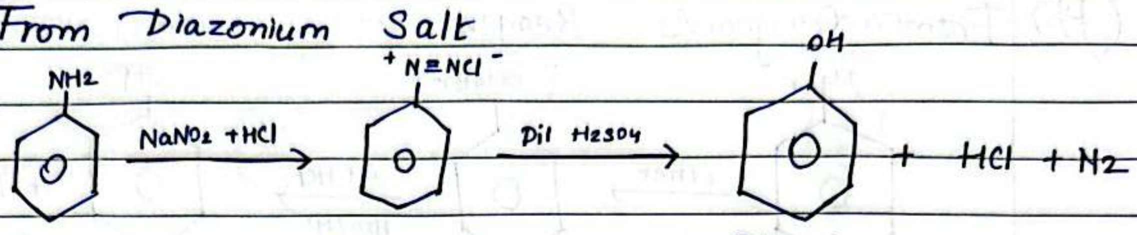
(b) From Benzene Sulphonic Acid.



Benzene   Benzene sulphonic acid   (Phenol)

/13/116

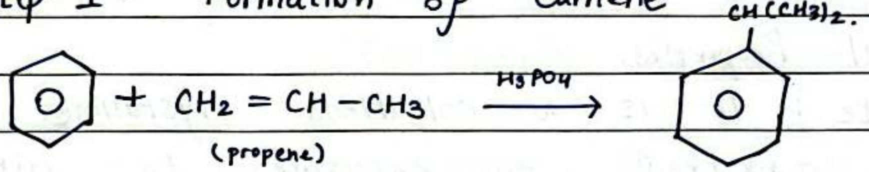
(c) From Diazonium Salt



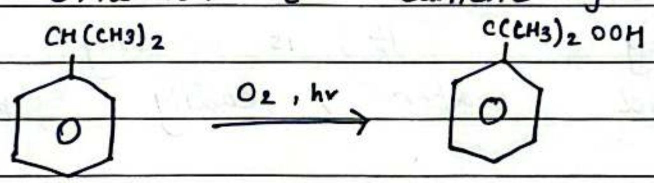
Aniline | Benzene diazonium chloride | Phenol.

(d) From Cumene — Industrial Process.

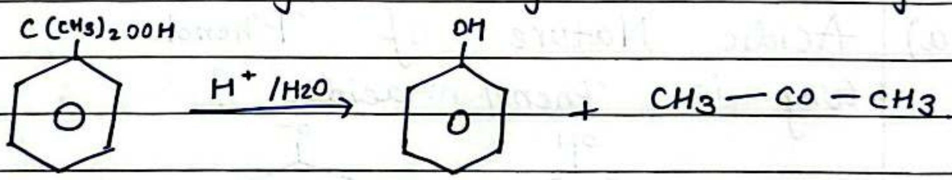
Step 1: Formation of Cumene



Step 2: Oxidation to cumene hydroperoxide



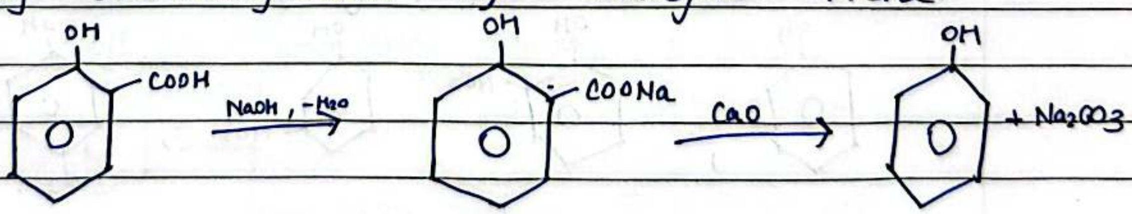
Step 3: Acid Catalysed Cleavage (Hock's rearrangement).



(Phenol) (Acetone).

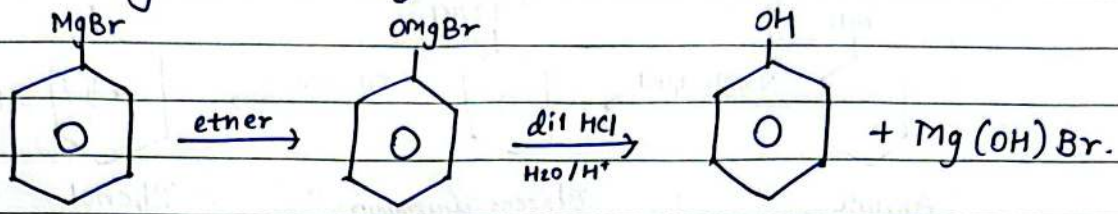
This is a major industrial process of Phenol.

(e) By decarboxylation of Salicylic Acid.



(Salicylic Acid) (Sodium salicylate) (Phenol)

(f) From Grignards Reagent.



(Phenyl magnesium Bromide) (Phenol)

⟹ Physical and Chemical Properties of Phenols.

- **Physical Properties.**

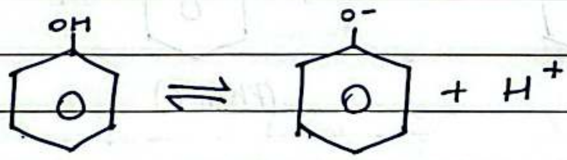
a) State : It is a colourless crystalline solid. turns pink/red on exposure to air with a sharp medicinal odour.

b) Solubility : It is only slightly soluble in cold water, readily soluble in organic solvents.

- **Chemical Properties of Phenol.**

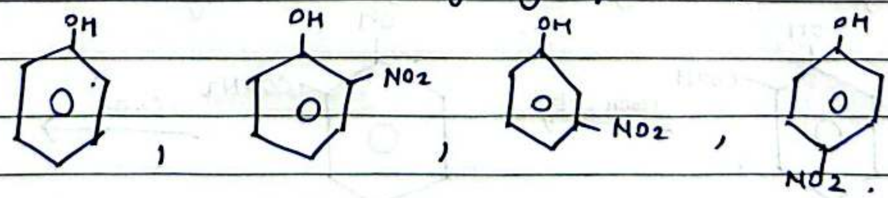
(a) Acidic Nature of Phenol.

Why is Phenol acidic ?



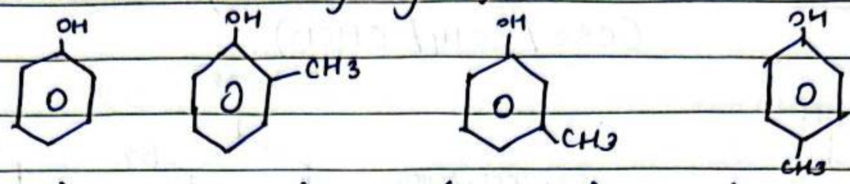
Phenoxide ion

i) Electron withdrawing groups (increase acidity).



Acidity : (d) p-nitrophenol ≈ (b) o-nitrophenol > (c) m-nitrophenol.

ii) Electron donating groups (decrease acidity).

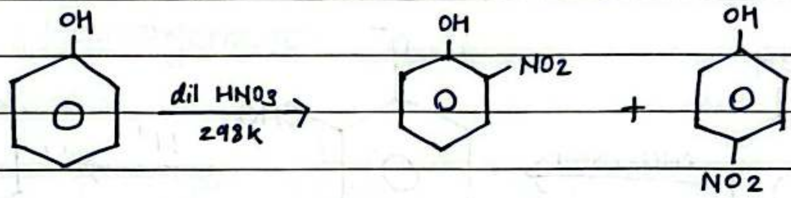


(Phenol) 1st (o-cresol) 4th (m-cresol) 2nd (p-cresol) 3rd.

Order of Acidity : (a) Phenol) > (m-cresol) > (p cresol) > (o-cresol).

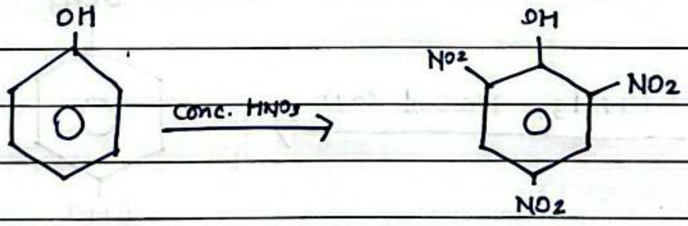
(b) Reaction of phenol due to benzene ring (Electrophilic Substitution).

1) Nitration : (a) with dilute HNO_3 at 298 K



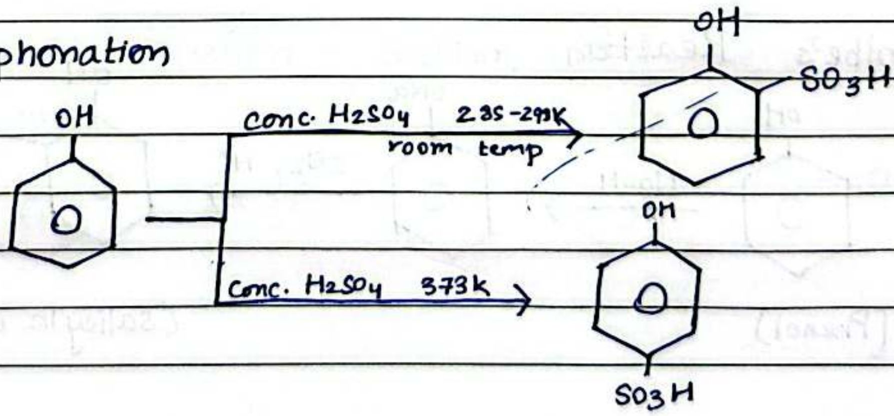
two isomers here can be seperated by steam distillation

(b) : with conc. HNO_3



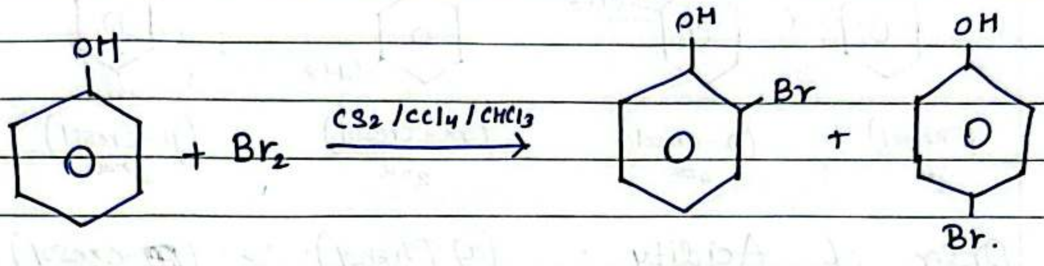
(2,4,6 trinitrophenol) Picric acid.

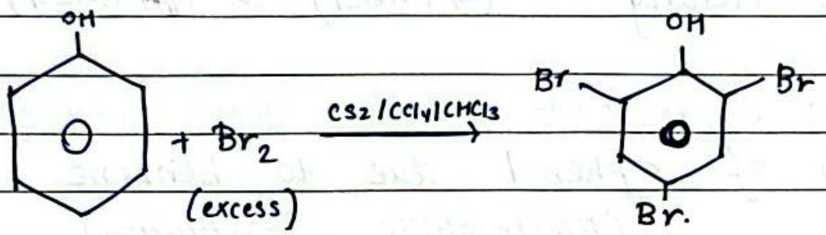
2) Sulphonation



__/16/119

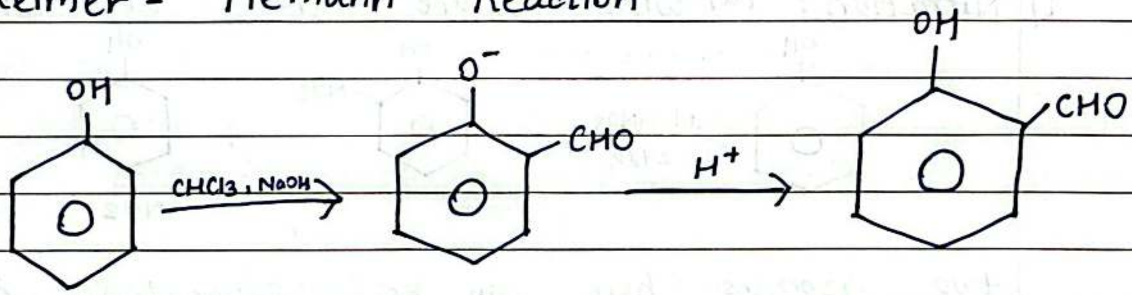
3) Halogenation : with a less polar solvent (CS_2 / CCl_4 / $CHCl_3$).





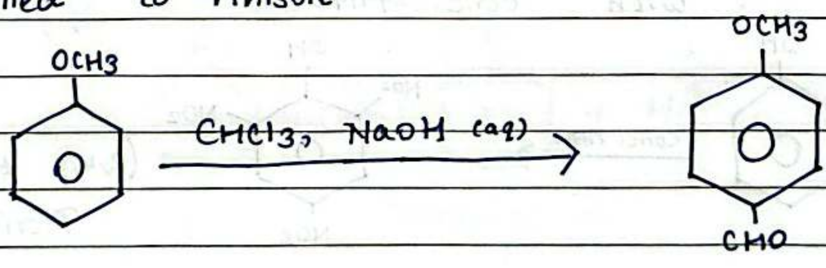
(c) Reimer - Tiemann Reaction

Imp. •



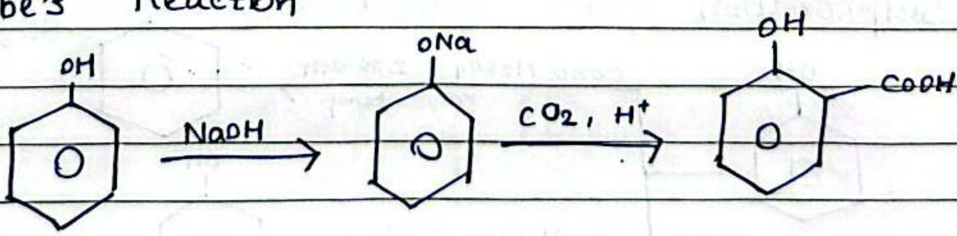
(Phenol) (Salicyaldehyde)

• Applied to Anisole



(p-Anisaldehyde)

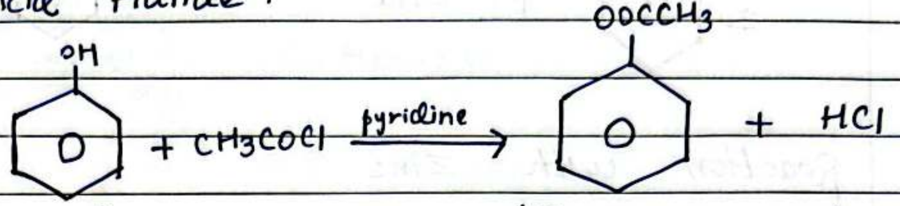
(d) Kolbe's Reaction



(Phenol) (salicylic acid).

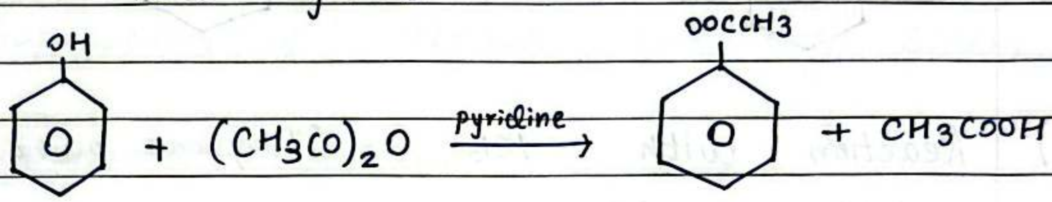
(e) Esterification of Phenol.

1) With Acid Halide :

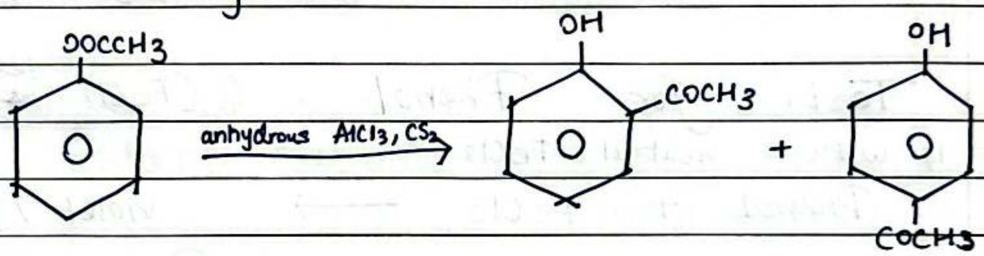


(Phenol) (Phenyl ethanoate)

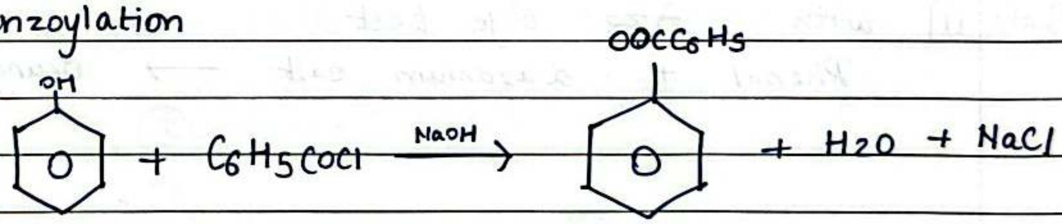
2) with Acid anhydride



3) Fries Rearrangement

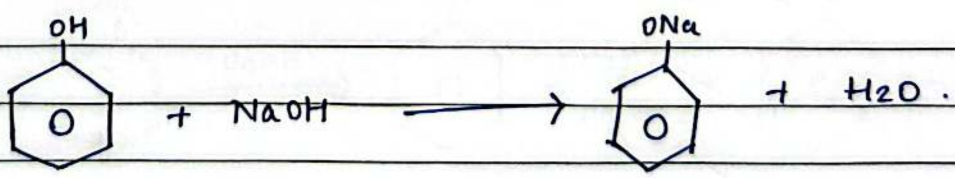


4) Benzoylation



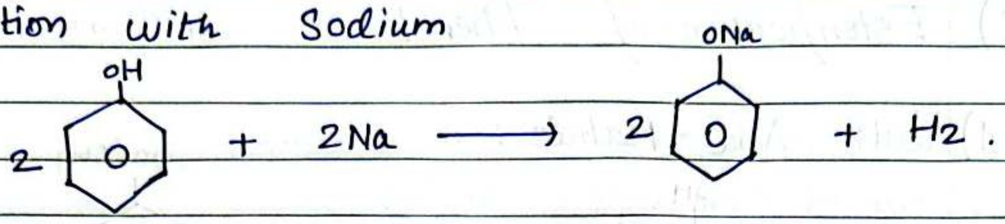
(Phenol) (Benzoyl chloride) (Phenyl Benzoate)

(f) Reaction with Sodium hydroxide

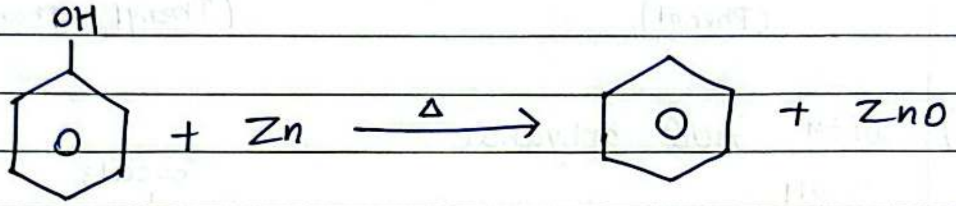


/18/121

(g) Reaction with Sodium



(h) Reaction with Zinc



(i) Reaction with PCl_5 (Phosphorus penta Chloride)

$$3\,C_6H_5OH + PCl_5 \longrightarrow (C_6H_5O)_3\,P{=}O + 3HCl$$

(j) Test for Phenol ($FeCl_3$ test)

i) with neutral $FeCl_3$

Phenol + $FeCl_3$ $\longrightarrow$ violet / purple colouration

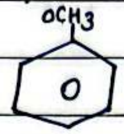
ii) with Azo dye test

Phenol + diazonium salt $\longrightarrow$ orange / red ppt.

⟹ Ethers — Structure & Nomenclature

Ethereal group : $R - O - R_2$

General group : $C_nH_{2n+2}O$

Example :  → Anisole

⟹ Preparations of Aliphatic and Aryl ethers

(a) Williamson's Synthesis

$$R - X + R - ONa \xrightarrow[\text{ether}]{\text{dry}} R - OR + NaX$$

→ Examples : ①

$$CH_3I + C_2H_5ONa \longrightarrow CH_3 - O - C_2H_5 + NaI$$

(methyl ethyl ether)

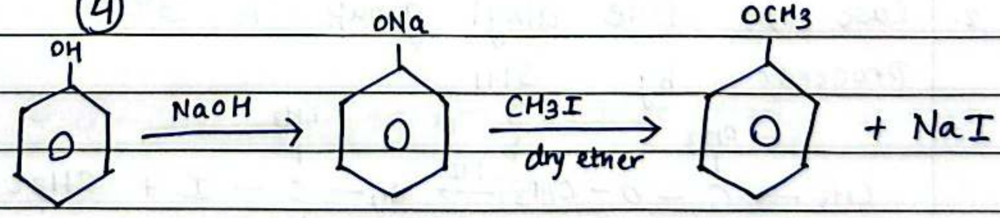
→ ②

$$C_2H_5Br + C_2H_5ONa \longrightarrow C_2H_5 - O - C_2H_5 + NaBr$$

→ ③

$$CH_3 - \underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}} - I + CH_3ONa \xrightarrow[\text{ether}]{\text{dry}} CH_3 - \underset{CH_3}{\underset{|}{C}} = CH_2 + CH_3OH + NaI$$

→ ④



__/20/123

(b) Dehydration of Alcohols

$$2\,C_2H_5OH \xrightarrow[140^\circ C \quad -H_2O]{conc\ H_2SO_4} C_2H_5 - O - C_2H_5$$

(dietnyl ether, ethoxyethane)

⟹ Properties of Ethers

(a) Reaction with HI (Cleavage of C-O bond)

Aliphatic Ether

$$R-O-R' + HI \longrightarrow R-I + R'OH.$$

Example : ①

$$C_2H_5 - O - C_2H_5 + HI \longrightarrow C_2H_5I + C_2H_5OH$$

②

$$CH_3 - O - C_2H_5 + HI \longrightarrow CH_3I + C_2H_5OH$$

Mechanism

1. Case (i) both alkyl groups are 1° : Reactions Proceed by SN2

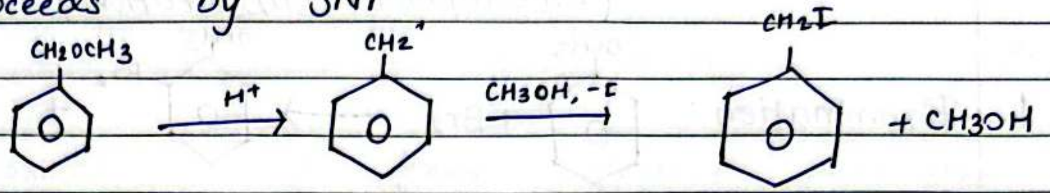
$$CH_3 - CH_2 - O - CH_2 - CH_3 \xrightarrow[SN_2]{HI} CH_3 - CH_2 - I + CH_3CH_2OH.$$

2. Case (ii) one alkyl group is 3° : Reaction Proceeds by SN_1

$$CH_3 - \underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}} - O - CH_3 \xrightarrow{HI} CH_3 - \underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}} - I + CH_3OH$$

/21/124

3. Case (iii) : One Alkyl Group is Benzyl : Reaction proceeds by S_N1



4. Case (iv) : One alkyl group is 2° : Then the outcome depends on solvent
— in protic solvent (H_2O or C_2H_5OH) by S_N1
— in acetone by S_N2.

(a) By H_2O (S_N1)

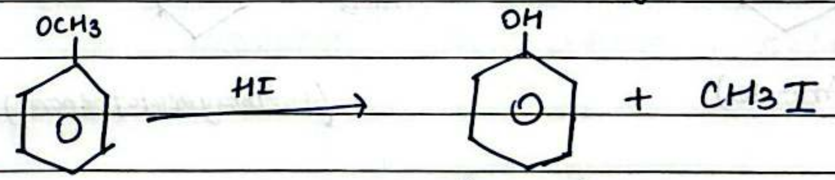
$$CH_3 - \overset{CH_3}{\overset{|}{CH}} - O - CH_3 + HI \xrightarrow{H_2O} CH_3 - \overset{CH_3}{\overset{|}{CH}} - I + CH_3OH$$

(b) By Acetone (S_N2)

$$CH_3 - \overset{CH_3}{\overset{|}{CH}} - O - CH_3 + HI \xrightarrow{Acetone} CH_3I + CH_3 - \underset{CH_3}{\underset{|}{CH}} - OH.$$

(a) Reaction with HI (Aryl Ether).

Imp



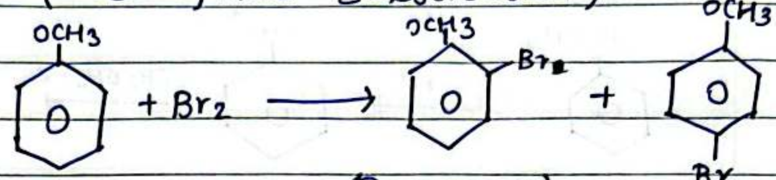
(Phenol)

with excess HI

$$CH_3 - \overset{CH_3}{\overset{|}{\underset{CH_3}{\underset{|}{C}}}} - O - CH_3 \xrightarrow[excess]{2HI} CH_3 - \overset{CH_3}{\overset{|}{\underset{CH_3}{\underset{|}{C}}}} - I + CH_3I + H_2O.$$

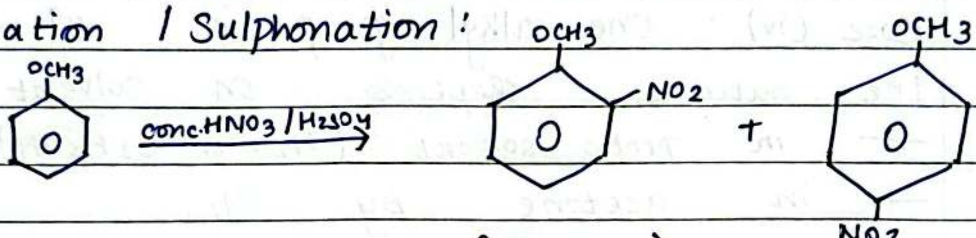
(b) Reactions of Aryl Ethers due to Benzene ring (Electrophilic Substitution)

1. Bromination :



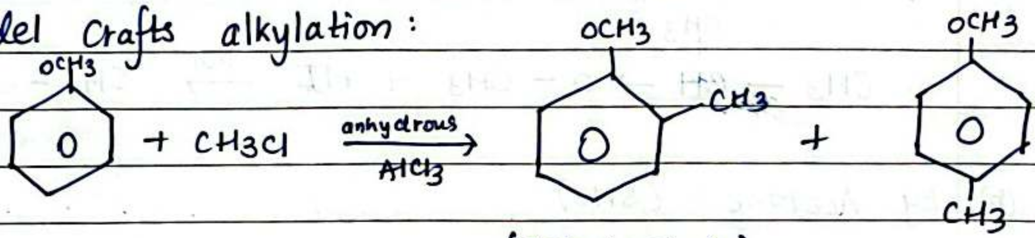
(Bromoanisole)

2. Nitration / Sulphonation :



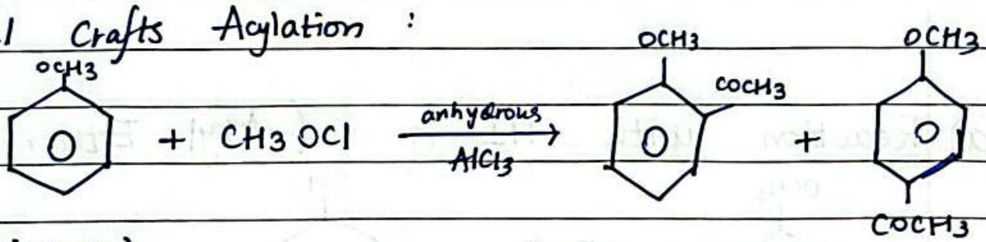
(nitroanisole)

3. Friedel Crafts alkylation :



(Methylanisole)

4. Friedel Crafts Acylation :



(Anisole) (methoxyacetophenone)

(c) Reaction with Chlorine :

$$C_2H_5OC_2H_5 + 2Cl_2 \xrightarrow{dark} CH_3Cl - O - \underset{}{\overset{Cl}{\overset{|}{C}H}} - CH_3 + 2HCl.$$

$$C_2H_5OC_2H_5 + 10Cl_2 \xrightarrow{sunlight} C_2Cl_5OC_2Cl_5 + 10HCl$$

(d) Oxidation (peroxide formation)

$$R-CH_2-O-CH_2-R + O_2 \xrightarrow{air, light} R-\underset{}{\overset{OOH}{\overset{|}{C}H}}-O-CH_2-R.$$

Autoxidation $\Longrightarrow$ forms explosive peroxides.

(e) Reaction with PCl_5.

$$R-O-R_2 + PCl_5 \longrightarrow R-Cl + R_2-Cl + POCl_3.$$

$\Rightarrow$ Physical Properties of Aliphatic and Aryl Ether.

i] Aliphatic ethers:

State : lower ethers (dimethyl ether) are gases
common ethers are volatile, only slightly soluble.
Miscible : with most organic Solvents.

ii] Aryl Ethers.

state : Anisole and Similar ethers are colourless liquids with pleasent odour.
practically insoluble in water, soluble in organic Solvents.

$\Rightarrow$ Uses of Ethers.

Diethyl ether was generally used as a general anaesthetic and as laboratery solvents.

Aryl ethers such as anisole are used as intermediates in perfumes.